DSLab. 07 Karnaugh Map

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- Design and verify the following circuits using Verilog HDL and Schematic
- Verilog
  - Behavioral level modeling
  - Dataflow modeling
  - Structural level modeling
- Schematic
- Please write and upload the lab report (Lab07) -- Due on 2023/10/20 23:59

Exercise 1

- $F(w,x,y,z) = w'yz' + w'xy + wxz + xyz + wx'y'$
- Draw the Karnaugh map and find all the simplest sum-of-products of F
- If the simplest sum-of-products of F are $F1$, $F2$ and $F3$, verify $F1 = F2 = F3$ (using Verilog Dataflow modeling)

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Exercise 2

- Draw the Karnaugh map of $F1(v, w, x, y, z) = \sum(1, 2, 3, 6, 7, 9, 10, 11, 14, 15, 17, 18, 19, 22, 25, 26, 27, 30)$ and find the simplest sum-of-products of this Boolean function.
- If the simplest sum-of-products of *F1* is *F2*, verify *F1* = *F2* (using Verilog Structural level modeling)

v=0

wx \ yz	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

v=1

wx \ yz	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

Exercise 3: Nine's Complement Converter (1/2)

- Nine's Complement Converter

Digits		Nine's Complements	
Decimal	BCD	Decimal	BCD
	$x_3\,x_2\,x_1\,x_0$		$y_3\,y_2\,y_1\,y_0$
0	0 0 0 0	9	1 0 0 1
1	0 0 0 1	8	1 0 0 0
2	0 0 1 0	7	0 1 1 1
3	0 0 1 1	6	0 1 1 0
4	0 1 0 0	5	0 1 0 1
5	0 1 0 1	4	0 1 0 0
6	0 1 1 0	3	0 0 1 1
7	0 1 1 1	2	0 0 1 0
8	1 0 0 0	1	0 0 0 1
9	1 0 0 1	0	0 0 0 0

Nine's-complement table

Exercise 3: Nine's Complement Converter (2/2)

- Design the nine's complement converter using Karnaugh maps and don't-care conditions, draw the logic diagram of the nine's complement converter and verify the circuit (using Verilog Structural level modeling)